



Aravindakshan R R 250701066-2025-CSE ▾

A2

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	g	t	a	b	
s2		g	x	t	x	a	y	b

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab	2
azb	

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char s1[1000], s2[1000];
6      scanf("%s", s1);
7      scanf("%s", s2);
8      int n = strlen(s1);
9      int m = strlen(s2);
10     int dp[n + 1][m + 1];
11     for (int i = 0; i <= n; i++)
12         dp[i][0] = 0;
13     for (int j = 0; j <= m; j++)
14         dp[0][j] = 0;
15     for (int i = 1; i <= n; i++) {
16         for (int j = 1; j <= m; j++) {
17             if (s1[i - 1] == s2[j - 1]) {
18                 dp[i][j] = dp[i - 1][j - 1] + 1;
19             }
20             else {
21                 dp[i][j] = (dp[i - 1][j] > dp[i][j - 1])
22                     ? dp[i - 1][j]
23                     : dp[i][j - 1];
24             }
25         }
26     }
27     printf("%d", dp[n][m]);
28     return 0;
29 }
```

Check

	Input	Expected	Got	
✓	aab azb	2	2	✓
✓	ABCD ABCD	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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